

VR Gaze prediction and data collection for children affected by learning deficiencies

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Figure 1: Snapshot from the Laboratory Scene illustrating the gaze prediction indicator.

Abstract

Assessing neurodevelopmental learning deficiencies in children can benefit greatly from immersive Virtual Reality (VR) environments. The downside being that clinical-grade VR eye-tracking relies on expensive dedicated hardware, while existing software-only gaze prediction models (such as DGaze) suffer from the lack of initiative to embed them into interactive scenes with a clearly defined scope, be it for medical or entertainment use.

CCS Concepts

• **Human-centered computing** → **Virtual reality**; **Interaction techniques**; • **Applied computing** → **Health informatics**; • **Computing methodologies** → **Neural networks**.

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1 Introduction

Neurodevelopmental learning deficiencies, such as Attention-Deficit/Hyperactivity Disorder (ADHD) and Autism Spectrum Disorder (ASD), negatively impact how children process spatial environments, prioritize visual information or sustain attention. Assessing these cognitive micro-behaviors has relied on 2D screen-based continuous performance tasks, which lack the credibility of real-world environments. Immersive VR scenes offer a paradigm shift in both diagnostics and therapeutic intervention, allowing clinicians to measure visual search efficiency, distractibility, and the ability to concentrate within controlled, dynamic 3D spaces.

2 Related works

Capturing attention metrics in VR presents a noticeable hardware barrier. Clinical-grade VR eye-tracking systems require expensive dedicated internal cameras, severely limiting the accessibility of these diagnostic tools for standard clinical practices/ public schools. Software-only gaze prediction models, such as the DGaze model,

offer a promising alternative by using head kinematics and the spatial positioning of objects to predict visual attention. The DGaze model was the starting point of our work, as it was open source and had an intuitive design. Its initial paradigm was to predict the user's gaze based on the closest objects to the camera, as long as they were situated on screen. This is a brilliant approach in sparse scenes. On the other hand, their research stated that 98.7% of gaze data collected lies in the central region of the screen, so we decided to favor that characteristic over the original distance from the camera for prioritizing candid objects.

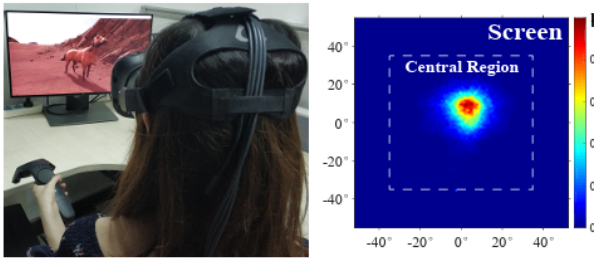


Figure 2: Snapshot from the DGaze paper illustrating gaze data.

2.1 Software-Based Gaze Prediction in VR

- **Head Kinematics and Visual Saliency:** Recent frameworks aimed to predict gaze in VR by fusing Head-Mounted Display (HMD) motion signals with visual saliency cues derived from video frames, which effectively eliminates the need for direct eye-tracking hardware, but capturing saliency features in real time is a tedious task.
- **Deep Learning Architectures:** The field of gaze estimation is actively shifting from rule-based mathematical systems to data-driven deep learning architectures, utilizing Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) to learn complex eye movement patterns. Most of these models rely on Long Short-Term Memory (LSTM) which can be best described as interlocked outputs.

2.2 VR Environments for Assessing ADHD and Attention

- **Simulation of Reality:** VR technologies can mitigate the limitations of traditional laboratory methods by providing a more relevant experience.
- **The VR Classroom:** Studies have successfully utilized VR classrooms where children perform math, Stroop, or continuous performance test (CPT) tasks while real-world distractors are introduced.
- **Attentional Disruption:** Eye-tracking within these VR contexts has speculated that while children with ADHD might not look at distractors for long periods, their presence disrupts "on-task gaze," making it hard for them to return to tasks once interrupted.

2.3 Gaze Tracking as a Biomarker for Autism

- **Oculomotor Differences:** Systematic reviews suggest that precise quantification of oculomotor behaviors, such as fixation duration, saccadic movements, and gaze patterns suggest that there is indeed a difference in attentional and cognitive processes between autistic children and typically developing peers.
- **Visual Target Preferences:** Research highlights distinct abnormalities in visual fixation patterns among autistic individuals, notably a reduced attention to social areas and an increased attention to unusual objects.

2.4 The Advantage of Adaptive Environments

- **Tailored Stimuli:** VR platforms are highly effective for evaluating sensory and executive function deficits because they provide a realistic experience.
- **Increased Cooperation:** VR environments can be created to the taste of children with developmental disabilities, which increases cooperation and is more likely to accurately reflect their abilities and sensory impairments, as they find themselves in an enjoyable place.

3 Eye Tracking without dedicated hardware

High-end Eye Tracking technology relies on infrared cameras to detect the pupil's location. These type of cameras are expensive and require calibration for each new user, limiting clinical accessibility. To bypass this hardware bottleneck, we adapted a software-only gaze prediction framework, significantly modifying its target prioritization logic to suit dense scenes and gamified tasks. The foundational DGaze model operates on the principle that head kinematics and visual saliency are highly reliable predictors of human gaze in immersive environments. Rather than tracking the physical pupil, the model utilizes a deep neural network to map head rotation, position, and velocity to 2D gaze coordinates. While this approach successfully eliminates the need for internal cameras, the original DGaze model was trained exclusively on HTC Vive hardware. Deploying this pre-trained model directly onto standalone headsets introduces severe spatial drift due to differing Field of View (FOV) parameters, rendering the raw output too erratic for concise medical telemetry.

3.1 Scene Implementation and Pipeline

To ensure high realistic validity, we designed the primary testing environment as a gamified virtual laboratory. The integration of the baseline gaze prediction framework is handled through three primary functional components:

- **The Client Module:** acts as the synchronous receiver for the 2D spatial coordinate outputs generated by the PyTorch prediction model.
- **The Dynamic Objects Array:** represents a continuous evaluation algorithm that identifies and isolates the three most contextually probable target objects within the user's current FOV.
- **The Data recorder:** is a telemetry module responsible for packing the spatial coordinates (x, y, z) of the isolated objects, alongside the user's head pitch and yaw angular velocities,

into a 50-frame rolling buffer. This kinematic buffer is subsequently transmitted as the input tensor to the PyTorch network.

3.2 Interactive and Administrative Components

To bridge the gap between the virtual environment and the supervisor, two administrative architectures were developed:

- **The Experiment manager:** This component establishes a real-time, bidirectional connection between the supervisor's dashboard and the user's VR application. It tracks the active task parameters and permits the supervisor to dynamically adjust difficulty levels or visual clues. Additionally, it streams behavioral metrics back to the supervisor in real-time, such as the duration of sustained gaze prior to an object selection.
- **The Experiment Item Protocol:** A modular labeling script applied directly to 3D assets within the scene. It provides an intuitive mechanism for assigning multiple categorical tags to interactive objects (e.g., categorizing a specific 3D model simultaneously as "chair" and "black"), thereby facilitating precise data querying and dynamic task generation.

3.3 Gaze Visualization and Adaptive Scaffolding

To translate the PyTorch model's numerical predictions into interactive gameplay mechanics, a dedicated gaze interaction and filtering logic was implemented within the Unity Client. First, a continuous raycast is projected from the virtual camera toward the 2D screen coordinates supplied by the DGaze model. To accurately detect object intersections, a physical spherical indicator (utilizing a Unity Rigidbody) is positioned along this ray. A critical implementation detail involved applying an inverted bitmask $\sim(1 \ll 2)$ to the raycast logic; this prevents the gaze indicator sphere itself from occluding the physical targets and disrupting the spatial distance calculations required by the neural network. When the predictive ray intersects with an active task target, a humble two-stage temporal filter is engaged to govern object selection: The system mandates a 20ms continuous hardware dwell time to confirm intentionality and filter out saccadic noise. Once the anti-jitter threshold is met, a secondary timer begins. The user must sustain visual fixation on the target for a defined duration (defaulting to 2.0 seconds) before selection is unlocked. To compensate for the natural micro-saccades of the human head during prolonged fixation, we implemented an angular "sticky" logic. If the user's predicted gaze drifts off the physical collider of the target, the system maintains the target lock provided the gaze remains within a forgiving 8-degree angular radius of the object's center. The system incorporates an adjustable, proximity-based visual scaffolding mechanism. Active task objects situated within a specific angular radius (e.g., 15 degrees) of the user's predicted gaze receive a dynamic gradient outline. The opacity of this secondary highlight scales inversely with the angular distance, such as to gently guide the user's visual search process without explicitly forcing attention.

3.4 Object validation and prioritization

Instead of calculating 3D distance thresholds, the Unity client projects all interactive objects from 3D world space into a normalized 2D screen space using the camera's projection matrix. The

system continuously calculates the distance of these projected coordinates to the absolute center of the user's field of view. The algorithm isolates and heavily favors the three objects positioned closest to the 2D screen center, entirely overriding their physical proximity to the camera. This shift from world-space depth to screen-space centrality anchors the predictive model to the user's natural focal point. By feeding the AI only the most contextually relevant targets, we drastically reduce computational overhead and establish a stable baseline. To validate that an object situated in the center of the screen is actually visible to the user we send a raycast to the target object. If the raycast hits the object it is good to go to the top of the dynamic objects array, if not, that means the target object is physically behind a larger one and the search for other candidates continues.

4 Spatial Mathematics and Clinical Metrics

To transition the VR environment from a simple interactive simulation into a potential diagnostic tool, the underlying logic relies heavily on continuous spatial calculations and temporal tracking.

4.1 Screen-Space Target Prioritization

As established in Section 3, the *Dynamic Object Array* prioritizes targets based on their proximity to the user's foveal center rather than their physical 3D distance. To achieve this, every active physical object O_i located at world coordinates (x_i, y_i, z_i) is continuously projected into 2D screen space using the VR camera's projection matrix.

Let the projected 2D coordinates of the object be $P_i = (u_i, v_i)$ and the absolute center of the headset's screen resolution be $C = (u_c, v_c)$. The pixel-space Euclidean distance D_i between the object and the visual center is calculated as:

$$D_i = \sqrt{(u_i - u_c)^2 + (v_i - v_c)^2} \quad (1)$$

To account for the physical architecture of the headset, this 2D pixel distance is then converted into an angular offset θ_i relative to the camera's forward vector. The system actively isolates the subset of objects where θ_i is minimized, ensuring that the PyTorch model is only fed data for targets actively occupying the user's central field of view.

4.2 Derivation of Diagnostic Metrics

The primary objective of the administrative dashboard is to translate the raw temporal data captured by the raycast into valuable clinical metrics. We formalize two critical biomarkers for assessing neurodevelopmental deficiencies:

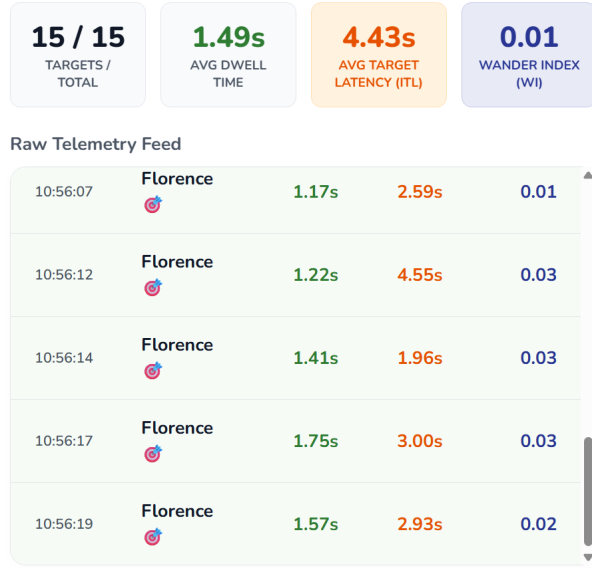
1. Inter-Target Latency (ITL): ITL measures the efficiency of a child's visual search algorithm. It is defined as the total time elapsed between the successful selection of the previous target and the selection of the current target. If t_k represents the exact timestamp when the k -th target is successfully clicked, the ITL for that specific target is:

$$ITL_k = t_k - t_{k-1} \quad (2)$$

By calculating the mean $\overline{ITL}$ over a full clinical session consisting of N targets, clinicians can establish a baseline for the child's processing speed and spatial navigation efficiency:

Session Analysis: P-001

2026-07-09 10:55 | Status: COMPLETED | Baseline Delay: 1s

**Figure 3: Test Run taken by the author**

$$\overline{ITL} = \frac{1}{N-1} \sum_{k=2}^N (t_k - t_{k-1}) \quad (3)$$

2. The Wander Index (WI): Children affected by ADHD frequently show signs of high distractibility, fixating on non-essential environmental stimuli. To evidetiate this, the system independently tracks the cumulative dwell time the user spends looking at active task targets (T_{task}) versus the time spent looking at background distractors ($T_{distractor}$).

The Wander Index (WI) is calculated as the ratio of distractor fixation to total fixation time, yielding a normalized score between 0 and 1:

$$WI = \frac{T_{distractor}}{T_{distractor} + T_{task}} \quad (4)$$

A WI approaching 0 indicates hyper-focus and sustained attention, while a higher WI suggests severe distractibility and a fragmented visual search pattern. By dynamically logging these values via the *Experiment Manager*, supervisors can quantitatively track a child's cognitive progression over multiple VR sessions.

5 Clinical Application and Use Cases

The integration of the Unity VR client with the local PyTorch backend establishes a robust spatial telemetry pipeline. By abstracting the raw kinematic data into accessible interfaces, the pipeline serves as a versatile tool for both clinical diagnostics and special education support.

5.1 Configurable Clinical Parameters

A core advantage of this software oriented approach is its adaptability. Children with neurodevelopmental deficiencies present a wide spectrum of cognitive and motor capabilities.

To address this, the system features a suite of adjustable parameters. The physics and temporal constraints of the VR gaze interaction are strictly managed by the Unity *Client Module*, as these variables are tightly coupled to the real-time raycast logic. However, the exact values for these constraints are passed from the clinician's dashboard.

The system features a developer-mode inventory cache. When a new 3D scene is loaded, the system automatically scans and saves all interactable objects to a persistent cache. This allows the clinician to dynamically generate specific visual search tasks in real-time. The primary configurable variables exposed to the clinician are detailed in Table 1.

Table 1: Configurable Clinical Parameters governing VR interactions.

Parameter	Clinical Function
Task Configuration	Defines the exact target objects, required quantity, and sequence order.
Sustained Attention Delay	Mandates a prolonged foveal fixation duration (e.g., 2.0s) on a target before selection is unlocked.
Retention Radius	Compensates for natural saccadic drift by maintaining target lock within a specific angular threshold (e.g., 8°).
Scaffolding Radius	Triggers proximity-based dynamic gradient outlines to assist and guide the user's visual search (e.g., 15°).

5.2 The Clinician's Telemetry Dashboard

To operationalize the mathematical metrics defined in Section 4, a secure, full-stack web dashboard was developed. The architecture utilizes a Flask-based RESTful API communicating with a relational MySQL database, ensuring that patient sessions are securely archived.

The frontend, built with React, acts as the primary administrative interface. It is divided into three core functional pages:

- **Patients Page:** A database management interface where educators and clinicians can add, edit, or track individual patient profiles and diagnosis notes.
- **Start Session Page:** The active control center where the clinician configures the parameters outlined in Table 1, selects the VR scene, and pushes the customized level directly to the Unity headset.

- **Session Archive:** The analytical review interface. Upon completion, it aggregates the telemetry to calculate the Average Target Latency (ITL) and the Wander Index (WI), alongside a chronological feed of every visual interaction.

5.3 Teaching Support and Gamified Interventions

Beyond diagnostics, the pipeline serves as a powerful assistive tool for special education teachers. Because the system utilizes standard commercial standalone headsets (e.g., Meta Quest), it can be easily deployed in classrooms.

Teachers can utilize the *Start Session* page to design gamified interventions. If a child struggles with severe distractibility, the teacher can decrease the *Sustained Attention* parameter and increase the *Scaffolding Radius*. This artificially lowers the cognitive load. As the child's spatial focus improves, the teacher can systematically tighten these parameters, effectively using the VR environment as a digital cognitive prosthetic.

6 Future Work and System Extensions

While the current pipeline successfully captures and categorizes visual attention telemetry, our roadmap includes two major system extensions aimed at enhancing both the underlying prediction architecture and the clinician's analytical capabilities.

6.1 Implicit Labeling for Predictive Correction

Currently, the system relies on the foundational DGaze model to predict base coordinates, utilizing an 8-degree angular *Retention Radius* to compensate for natural head jitter. In future iterations, we intend to completely phase out this dependency.

To achieve this, we developed a "Task-Driven Ground Truth" (or Implicit Labeling) paradigm within our Unity environment. Traditional machine learning for gaze tracking requires laborious manual annotation of where a user is looking. Instead, our system forces natural gaze behavior by explicitly instructing the user to "Look at the Red object and pull the trigger." Because the system inherently knows the spatial coordinates (x, y, z) of the designated target before the interaction occurs, the need for manual data labeling is entirely eliminated.

A critical challenge in this paradigm is distinguishing the "Searching" phase (saccadic movement) from the "Fixating" phase. To capture uncontaminated data, the *Data Recorder* maintains a continuous rolling buffer of the last 50 frames of kinematic data. The moment the user pulls the controller trigger, the system extracts this exact pre-click buffer, tags it with the target's spatial coordinates, and archives it.

By compiling thousands of these localized "Hit/Miss" kinematic windows, we are generating a highly robust dataset. This pure regression data will be used to train a custom, lightweight, hardware-agnostic spatial correction model built entirely upon Quest hardware data, with the hopes of pushing the prediction accuracy beyond the current baseline.

6.2 Visual Search Efficiency and Path Tracking

To provide clinicians with deeper insights into a child's spatial navigation strategy, we are developing a post-session path tracking

visualizer. When instructed to locate a specific target, a neurotypical brain generally employs a systematic and logical visual search pattern. In contrast, children with learning or developmental difficulties often exhibit highly chaotic and erratic visual scanning behaviors.

Rather than relying solely on temporal metrics such as the Wander Index, this future tool will quantify and visualize the spatial efficiency of the user's gaze. The system will calculate the total angular distance the predicted raycast travels across the environment before successfully resting on the target object. A short, direct gaze path indicates high search efficiency and cognitive control, whereas a long, erratic path suggests that the child is overwhelmed by the surrounding visual stimuli.

The dashboard will map this continuous raycast as a color-coded spline through a miniature 3D representation of the laboratory scene. This will allow educators and doctors to replay the session and qualitatively differentiate between a child who systematically scans a room from left-to-right, versus one whose gaze bounces aimlessly, providing a profound spatial context to the quantitative data.

7 Conclusion

This paper presented a novel, hardware-agnostic spatial telemetry pipeline designed to assess and train visual attention in children affected by neurodevelopmental deficiencies. By successfully adapting a software-only gaze prediction model and inverting the target prioritization logic from 3D world-space to 2D foveal-space, we eliminated the need for expensive, clinical-grade eye-tracking hardware.

To conclude with, the mathematical derivation of the Inter-Target Latency (ITL) and the Wander Index (WI), coupled with a full-stack, secure clinician dashboard, elevates this VR environment from a simple simulation into a potential diagnostic instrument. As virtual reality continues to penetrate standard classroom and clinical settings, systems like the one proposed in this research will serve as highly accessible digital cognitive prosthetics, offering new avenues for the diagnosis and therapeutic intervention of attentional disorders.

References

- [1] Zhiming Hu, Sheng Li, Congyi Zhang, Kangrui Yi, Guoping Wang, and Dinesh Manocha. 2020. DGaze: CNN-Based Gaze Prediction in Dynamic Scenes. *IEEE Transactions on Visualization and Computer Graphics*, 26, 5 (2020), 1902–1911.
- [2] Christos Petrou, Harris Partaourides, Athanasios Balomenos, Yannis Kopsinis, and Sotirios Chatzis. 2023. Gaze Prediction in Virtual Reality Without Eye Tracking Using Visual and Head Motion Cues. *IEEE Access*, 11 (2023), 29424–29436.
- [3] Vincent Sitzmann, et al. 2018. Saliency in VR: How do people explore virtual environments?. *IEEE Transactions on Visualization and Computer Graphics*, 24, 4 (2018), 1633–1642.
- [4] Vivian Clay, et al. 2019. Eye tracking in virtual reality. *Journal of Eye Movement Research*, 12, 1 (2019).
- [5] Mariano Alcañiz, Javier Marin-Morales, et al. 2021. Eye gaze as a biomarker in the recognition of autism spectrum disorder using virtual reality and machine learning. *Autism Research*, 14, 7 (2021), 1436–1449.
- [6] Jared D. Stokes, et al. 2022. Measuring Attentional Distraction in Children With ADHD Using Virtual Reality Technology With Eye-Tracking. *Frontiers in Virtual Reality*, 3 (2022), 855895.
- [7] Alain Neguț, Andreea M. Jurma, and Daniel David. 2017. Virtual-reality-based attention assessment of ADHD: ClinicaVR: Classroom-CPT versus a traditional CPT. *Child Neuropsychology*, 23, 4 (2017), 472–492.
- [8] Albert A. Rizzo, et al. 2000. The Virtual Classroom: A Virtual Reality Environment for the Assessment and Rehabilitation of Attention Deficit Hyperactivity Disorder.

CyberPsychology and Behavior, 3, 3 (2000), 483–499.